The white squares of a board: OEIS A230650

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Abstract

OEIS A230650 carries an empirical recurrence of order 23 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The entry's condition is decided by a bounded amount of state carried down the array; the admissible windows are the vertices of a finite digraph and the arrays counted are exactly the walks in it. Hence $a(n)$ is a walk count on $S = 30$ vertices and is C-finite, and whether the conjectured recurrence annihilates it is settled by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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1 The conjecture

OEIS A230650 is "Number of nX6 0..2 white square subarrays x(i,j) with each element diagonally or antidiagonally next to at least one element with value 2-x(i,j).". It has offset 1 and begins

$$0, 33, 177, 2547, 28161, 337977, 3951657, 46564959, \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 8*a(n-1) +50*a(n-2) -22*a(n-3) -437*a(n-4) -1133*a(n-5) -548*a(n-6) +2623*a(n-7) +5757*a(n-8) +6227*a(n-9) -194*a(n-10) -2664*a(n-11) -3827*a(n-12) +2327*a(n-13) -5991*a(n-14) -4452*a(n-15) -3865*a(n-16) -6535*a(n-17) -536*a(n-18) -639*a(n-19) -373*a(n-20) -29*a(n-21) +76*a(n-22) -16*a(n-23) for n>24

As of the "Last modified" line on the live entry (Jun 02 2025 08:43:59, revision 6) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The white squares are a board of their own

The board has 6 columns and grows one row at a time, but only the cells with $i + j$ even carry a value, from the alphabet $\{0, 1, 2\}$. The diagonal and antidiagonal neighbours of such a cell again have $i + j$ even, so the white squares form a board of their own and the entry's condition never refers to anything else: each white cell must have at least one neighbour, diagonally or antidiagonally, holding the value $2 - x$ where x is its own.

That looks one row up and one row down, so three consecutive rows settle the condition on the middle one. Carry two rows as the state, judge a row when the row after it arrives, and judge the row with nothing after it by the end vector. A cell at the edge of the board simply has fewer neighbours, and having none of the right value is a failure rather than a special case.

The white cells of a row occupy alternate columns, and which alternate ones depends on the parity of that row, so the two shapes alternate down the board; the state carries that parity in the shape of the row it holds.

Merging vertices with identical futures leaves $S = 30$. An admissible board is exactly a walk, so with M the adjacency matrix and ι, τ the vectors recording which states may begin and end a board,

$$a(n) \;=\; \iota^\top M^{n-1}\tau,$$

the exponent fixed by the entry's own shape and checked against its published terms. In particular a is C-finite.

3 The proof

Lemma 1. *Let $q(t) = t^{23} - \sum_i c_i t^{23-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+23} - \sum_i c_i M^{j+23-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. A constant factor in front of a divides out of the residual and changes nothing here. $\square$

Theorem 2.

$$a(n) \;=\; 8\,a(n-1)+50\,a(n-2)-22\,a(n-3)-437\,a(n-4)-1133\,a(n-5)-548\,a(n-6)+2623\,a(n-7)+5757\,a(n-8)+$$

for every $n > 24$.

Proof. The vector $w = q(M)\tau$ was formed by 23 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 24 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

4 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the reading was pinned before the digraph was written, by a program that enumerates every filling of the white cells one by one and tests, for each of them, whether some diagonal or antidiagonal neighbour holds the required value. That program shares no code with the transfer matrix, so an error in one does not hide an error in the other.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A230650.

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[3] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.